MIL-A-13550E(EA)
16 September 1974
SUPERSEDING
MIL-A-13550D(MU)
4 December 1969

MILITARY SPECIFICATION

ANTIFOGGING KIT, M1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of chemical-biological mask antifogging kit.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-10857 - Cloth, Antifogging.

STANDARDS

FEDERAL

FED-STD-123 - Marking for Domestic Shipping (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

B5-16-1 - Antifogging Kit, M1.

FSC 6850

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials listed on Drawing B5-16-1 shall conform to the specifications listed thereon and to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the M1 antifogging kit shall conform to the specifications and drawings listed on Drawing B5-15-1 and subsidiary drawings.

3.2 Manufacture and assembly. The kit shall be manufactured and assembled in accordance with Drawing B5-15-1.

3.3 Compatibility. After the antifogging kit has been conditioned in accordance with 4.3.3.1, the antifogging cloth shall not change color or adhere to the can and shall meet the antifogging properties (before aging) of MIL-C-10857. In addition, the can assembly shall show no evidence of deformation, pitting, corrosion, loss of marking, or other imperfections when tested as specified in 4.3.3.1.

3.4 Low temperature brittleness. The can and cap shall show no evidence of cracking when tested as specified in 4.3.3.2.

3.5 Leakage. The antifogging kit shall gain no more than 1.5 grams in weight when tested as specified in 4.4.4.1.

3.6 Preproduction. Prior to the start of regular production and following any change in methods, materials, or equipment, a preproduction sample of antifogging kits shall be produced in accordance with this specification for examination and tests (see 4.3).

3.7 Workmanship. The antifogging kit shall be free from damage such as cracks; dents; punctures; split seams; burred edges; chipped paint; raveled, torn, or discolored cloth; and from foreign matter (dirt, grease, or oil).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

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4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 125 antifogging kits shall be produced using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. The preproduction sample shall be examined in accordance with the classification of defects (4.4.3.2).

4.3.2.2 For tests. Fifty kits shall be tested for compatibility as specified in 4.3.3.1, twenty-five kits shall be tested for low temperature brittleness as specified in 4.3.3.2, and fifty kits shall be tested for leakage as specified in 4.4.4.1. The acceptance number is zero.

4.3.3 Tests.

4.3.3.1 Compatibility. The antifogging kit shall be placed in an oven at 160° ± 5°F for five days. The kit shall then be removed from the oven and allowed to cool at room temperature for a minimum period of 24 hours. The antifogging cloth shall be removed from the can and the cloth and the components of the can shall be examined for damage. The marking on the

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can shall be tested for durability by rubbing hard three times with the fingers. The cloth shall then be tested for antifogging properties in accordance with MIL-C-10857, except that the accelerated aging specified therein is not required.

4.3.3.2 Low temperature brittleness. The antifogging kits shall be subjected to -65° ± 5°F for 24 hours. At the end of the storage period and while still at that temperature they shall be dropped one at a time from randomly held positions and a minimum height of six feet onto a flat concrete or firm metal surface.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the M1 antifogging kits produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions. However, no more than one lot of antifogging cloths shall be represented in any one lot of finished antifogging kits.

4.4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedure.

4.4.3.1 Examination and tests. Sample antifogging kits shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 Classification of defects.

(a) Antifogging kit, M1 (Dwg B5-16-1).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 2.5 percent defective	
101	Component missing, incorrect, or incorrectly assembled	

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4.4.3.2 Classification of defects (Cont'd).

(a) Antifogging kit, M1 (Dwg B5-16-1).

Categories	Defects	Acceptance standards
Major:	AQL 2.5 percent defective	
102	Overall length incorrect	
103	Threads on can and cap do not engage properly	
104	Can inside or outside diameter incorrect, as applicable	
105	Interior coating missing, when applicable	
106	Marking missing, incorrect, or illegible	
107	Workmanship (see 3.7)	
108	Leakage	4.4.4.1
Minor:	AQL 4.0 percent defective	
201	External coating missing	
202	Knurling on cap missing, when applicable	
203	Color incorrect	

4.4.4 Tests.

4.4.4.1 Leakage. Test shall be conducted as follows: Remove the cap and weigh the can and contents to the nearest decigram. Replace the cap and immerse the antifogging kit in 12 ± 1 inches of water at 70° ± 5°F for 15 ± 1 minutes. Remove the antifogging kit from the water and wipe dry. Remove the cap, wipe the can threads dry, and reweigh the can and contents.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking, interplant shipment (see 6.4). The antifogging kit shall be packaged and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use of further processing. Shipping containers shall be in compliance with the rules and regualtions applicable to the mode of transportation. Marking shall be in conformance with FED-STD-123 and shall include the contractors name, address, and contract number.

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5.2 Repair parts. When this item is being procured for storage and issue as a repair part, preservation, packaging, packing, and marking shall be as specified on the packaging data sheet which is identified by its National Stock Number.

6. NOTES

6.1 Intended use. This specification covers an antifogging kit intended to be used to prevent fogging of chemical-biological mask lenses.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Internation interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 216 and 234. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.4 Interplant shipments. Packaging for supplies and materials which will not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6850-A537

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Commander
Edgewood Arsenal
ATTN: SAREA-TS-S
Aberdeen Proving Ground, MD 21010

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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